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Exercise 1.6

Required

Textbook p. 34

There are four batches of components. The first batch contains 2 000 components, of which 5% are defective. The second batch contains 500 components, of which 40% are defective. The third and fourth batches each contain 1 000 components, with approximately 10% defective. A batch is selected at random, and then a component is drawn at random from that batch. Find:

(1) The probability that the selected component is defective.

(2) The probability that the component came from the second batch, given that it is defective.

Answer / Working

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Exercise 1.14

Required

Textbook p. 34

The discrete random variables X and Y have the following joint probability mass function. Find:

$X\backslash Y$	-1	0	1
0	0.07	0.18	0.15
1	0.08	0.32	0.20

(1) The joint distribution function and joint probability density function of X and Y;

(2) The marginal probability mass functions of X and Y;

(3) The probability mass function of $Z = XY$;

(4) The correlation coefficient of X and Y.

Answer / Working

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Exercise 1.16

Required

Textbook p. 35

Let $X \sim N(0,1)$ and $Y \sim N(0,1)$ be independent random variables. Define $U = X + Y$ and $V = X - Y$.

(1) Find the joint probability density function $f_{UV}(u,v)$ of (U,V).

(2) Determine whether U and V are independent.

Answer / Working

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Exercise 1.21

Required

Textbook p. 35

For the random variables X and Y, it is known that

$$EX = 1, \quad EY = 3, \quad D(X) = 4, \quad D(Y) = 16, \quad \rho_{XY} = 0.5,$$

Let $U = 3X + Y$ and $V = X - 2Y$. Find EU, EV, $D(U)$, $D(V)$, and $\mathrm{Cov}(U, V)$.

Answer / Working

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Exercise 1.24

Required

Textbook p. 35

The random variable X is uniformly distributed on $[0, a]$. Given X, the random variable Y is uniformly distributed on $[X, a]$. Find:

(1) $E(Y \mid X)$, $0 \leq X \leq a$;

(2) EY.

Answer / Working

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Exercise 1.33

Required

Textbook p. 36

The joint characteristic function of the random variables X and Y is

$$\phi_{XY}(u,v) = \frac{6}{6 - 2\mathrm{j}u - 3\mathrm{j}v - uv}.$$

Find:

(1) The characteristic function of X;

(2) The expectation and variance of Y.

Answer / Working

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Exercise 1.9

Optional

Textbook p. 34

The random variable X has the following probability mass function:

X	1	2	3
P	0.2	0.5	0.3

Find the probability density function and distribution function of X, and sketch their graphs.

Answer / Working

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Exercise 1.11

Optional

Textbook p. 34

The random variable X has probability density function $f(x) = ae^{-|x|}$. Find:

(1) The coefficient a;

(2) Its distribution function.

Answer / Working

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Exercise 1.28

Optional

Textbook p. 36

The random variables X_1, X_2, X_3 are mutually independent, with characteristic functions $\phi_1(v)$, $\phi_2(v)$, and $\phi_3(v)$, respectively. Find the characteristic function of each of the following random variables:

(1) $X = X_1 + X_2$;

(2) $X = X_1 + X_2 + X_3$;

(3) $X = X_1 + 2X_2 + 3X_3$;

(4) $X = 2X_1 + X_2 + 4X_3 + 10$.

Answer / Working

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Exercise 1.29

Optional

Textbook p. 36

For each of the following characteristic functions of a random variable X, find its probability density function, mean, mean-square value, and variance.

(1) $\phi(v) = 0.2 + 0.3\mathrm{e}^{\mathrm{j}2v} + 0.2\mathrm{e}^{\mathrm{j}4v} + 0.2\mathrm{e}^{-\mathrm{j}2v} + 0.1\mathrm{e}^{-\mathrm{j}4v}$;

(2) $\phi(v) = 0.3\mathrm{e}^{\mathrm{j}v} + 0.7\mathrm{e}^{-\mathrm{j}v}$;

(3) $\phi(v) = \dfrac{4}{4 - \mathrm{j}v}$;

(4) $\phi(v) = \dfrac{\sin 5v}{5v}$.

Answer / Working

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Exercise 1.30

Optional

Textbook p. 36

Use the Fourier transform to derive the characteristic function of the uniform distribution.

Answer / Working